
PS Analysis in Data Science: Problem Set 12

Problem 1. Find the extrema of $f : \mathbb{R}^3 \rightarrow \mathbb{R}$,

$$f(x, y, z) = xyz$$

subject to the constraint

$$x^2 + y^2 + z^2 = 12.$$

Problem 2. Consider the ellipse

$$M = \{(x, y)^\top \in \mathbb{R}^2 : 4x^2 + y^2 = 16\}.$$

- (a) Determine the points that are closest to and farthest from the origin.
- (b) Visualize the problem in Julia.
- (c) Determine the points on the ellipse that are closest to and farthest from $(3, 0)^\top$.

Problem 3. Find the extrema of $f : \mathbb{R}^3 \rightarrow \mathbb{R}$,

$$f(x, y, z) = x^2 + y^2 + z^2$$

subject to the constraints

$$x + y + z = 6, \quad x - y = 2.$$

Problem 4. Consider the function $f : \mathbb{R}^2 \rightarrow \mathbb{R}$,

$$f(x, y) = x^2 - y^2$$

subject to the constraint

$$x^2 + y^2 = 4.$$

1. Generate a heatmap of f on the square $[-3, 3]^2$ and draw the constraint curve on top of the heatmap.
2. Parametrize the constraint by

$$\gamma(t) = (2 \cos t, 2 \sin t), \quad 0 \leq t \leq 2\pi.$$

and plot the function $F(t) = f(\gamma(t))$ with t on the horizontal axis.

3. Make an educated guess of the exact maxima and minima based on this investigation. Then determine the exact constrained extrema using Corollary 5.4.7.